Executa – OpenRouter AI Architecture & Model Allocation

This document defines the recommended OpenRouter model architecture for Executa. It is intended for the development team so they understand exactly which model should be used for each AI-powered feature, why it is selected, and how fallback routing should work.

1. Recommended Model Stack

Primary Model: Qwen 3
Reasoning Model: DeepSeek R1 Distill
Fallback Model: Llama 3.3 70B

OpenRouter acts as the routing layer and allows Executa to switch models without changing backend architecture.

2. Scope Generation

Model: Qwen 3
Purpose: Generate scope documents, functional units, included features, excluded features, complexity analysis, and project summaries.
Why: Strong structured output, excellent requirement understanding, and reliable document generation.

3. Custom Feature Expansion

Model: Qwen 3
Client submits a simple feature request. The AI expands it into a complete functional unit with description, requirements, complexity score, and business logic.

4. Scope Upgrade Generation

Model: Qwen 3
Used when a client requests additional functionality after project approval. The model generates a new functional unit, included features, complexity score, and project impact summary.

5. Freelancer Vetting Test Generation

Model: Qwen 3
Generates specialization-specific assignments, business context, technical requirements, deliverables, and constraints for freelancer testing.

6. Freelancer Match Analysis

Model: DeepSeek R1 Distill
Used for compatibility reasoning between project requirements and freelancer profiles. Produces detailed written justification explaining why a freelancer is or is not suitable.

7. Recommended Match Score Logic

Do not allow AI to generate the match percentage directly. Instead: Skills = Platform Calculation Experience = Platform Calculation Vetting Results = Platform Calculation Availability = Platform Calculation The platform computes the score. AI only generates the explanation. This creates consistency and avoids disputes.

8. OpenRouter Routing Strategy

Project Scope Generation → Qwen 3
Functional Units → Qwen 3
Custom Feature Expansion → Qwen 3
Scope Upgrades → Qwen 3
Freelancer Vetting Tests → Qwen 3
Freelancer Match Explanation → DeepSeek R1 Distill
Fallback for All Features → Llama 3.3 70B

9. User Capacity Guidance

MVP Testing: 10–50 active users/day
Beta Launch: 50–100 active users/day
Production Scale: Move to paid OpenRouter usage and monitor token consumption.

10. Final Recommendation

For Executa, the best OpenRouter architecture is: **Qwen 3 (Primary)** + **DeepSeek R1 Distill (Reasoning)** + **Llama 3.3 70B (Fallback)**. This combination provides the strongest balance of cost, quality, structured output, and scalability.